Shanshan Mao

PhD Researcher (Awaiting Viva)
Multi-Agent Systems and Artificial Life
School of Computer Science
University of Birmingham

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Research Profile

PhD researcher in Computer Science at the University of Birmingham; thesis submitted in August 2026, awaiting viva voce examination. Research focuses on dynamic multi-agent systems, artificial life, and complex adaptive systems, with particular emphasis on individual agency, emergent macro-level dynamics, evolutionary adaptation, and signal manipulation.

Education

University of Birmingham, UK *Mar 2021 – Present*
PhD in Computer Science
Thesis submitted in August 2026; awaiting viva.

- Research focuses on emergence, stability, and collapse of hierarchy in multi-agent systems under bounded rationality and deceptive signalling.

University of Birmingham, UK *Sept 2019 – Oct 2020*
MSc in Advanced Computer Science *Distinction*

- Selected modules: Machine Learning, Evaluation Methods and Statistics, Introduction to Human Computer Interaction, Neural Computation.

Henan University, China *Sept 2014 – Jun 2018*
BSc in Computer Science and Technology

- Final grade: 80.5/100

Publications

S. Mao, P. Tiňo: From Cooperation to Hierarchy: A Study of Dynamics of Hierarchy Emergence in a Multi-Agent System.
Applications of Evolutionary Computation (EvoApplications 2026), LNCS, vol. 16524, pp. 199–215, Springer, 2026.

S. Mao, P. Tiňo: When Deception Spreads: Transmissible Deception Destabilises Emergent Hierarchy in a Cooperative Multi-Agent System.
ALIFE 2026: Proceedings of the 2026 Artificial Life Conference, MIT Press, 2026.

Academic Presentations

Strategic Interaction and Emergent Hierarchy in Multi-Agent Systems
Lightning Talk, Game Theory in the Era of AI Workshop, 2026

How Hierarchies Emerge Without Leaders: Insights from Multi-Agent Systems
Presentation, Birmingham International Scholars Forum, Sub-Forum on Artificial Intelligence and Digitalization, 2026

How Does Hierarchical Structure Emerge in Decentralized Multi-Agent Systems
Presentation, Toulouse, France, Evostar Applications conference, 2026

Research Experience

PhD Researcher, University of Birmingham *Mar 2021 – Present*
Emergence of Hierarchy in Multi-Agent Systems

- Developed an agent-based simulation framework for studying hierarchy emergence under local interaction and cooperation.
- Investigated the formation, persistence, and collapse of hierarchical structures under deceptive signalling and evolutionary adaptation.
- Applied trophic incoherence and network-based metrics to quantify structural organisation and disorder.
- Conducted large-scale simulation experiments and statistical analyses across multiple parameter settings and random seeds.

Master's Project, University of Birmingham *Nov 2019 – Jun 2020*

Heuristic Optimisation for the Constrained Travelling Salesman Problem

- Investigated metaheuristic optimisation methods for constrained travelling salesman problems.
- Evaluated Genetic Algorithms, Tabu Search, and Simulated Annealing on benchmark optimisation tasks.

Teaching Experience

Postgraduate Teaching Assistant, University of Birmingham *Sept 2022 – Mar 2026*

- Delivered tutorials and lab sessions in Data Structures, Databases, AI/ML, and Evolutionary Computation.
- Provided technical guidance and assessment support for large undergraduate cohorts.
- Assessed coursework, examinations, and programming assignments for large undergraduate cohorts.

Teaching Assistant, Machine Learning Summer School *Jul 2025*

- Supported hands-on teaching of machine learning and deep learning frameworks (scikit-learn, PyTorch, TensorFlow).

Research Projects

AI Companion Robotics (Doctoral Innovation Programme) *Sept 2023 – Jun 2024*

- Led development of a multi-modal AI companion prototype integrating speech, vision, and interaction modules.

User Interface Developer (Project) *2018*

- Developed Python-based UI to interface with hardware simulation systems via API.

Skills

Programming: Python, C++, MATLAB, JavaScript
ML / Data: PyTorch, scikit-learn, NumPy, pandas, Matplotlib, Seaborn
Research Methods: Agent-based modelling, complex systems simulation, network analysis, statistical modelling, hypothesis testing, reinforcement learning environments
Tools: Git, Jupyter, VS Code, PyCharm, LaTeX

Awards

First Prize, Sustainable Future in the City Competition *Oct 2019*